

A woman with long brown hair and a young child with blonde hair are sitting at a white table. The woman is leaning forward, looking at the child with a focused expression. The child is also leaning forward, looking back at the woman. On the table in front of them are several small, colorful pieces of clay or dough in various shapes and colors like orange, green, yellow, pink, purple, and red. The background is a softly blurred indoor setting with a plant and some furniture.

“Mommy,
where do
scientific
journal articles
come from?”

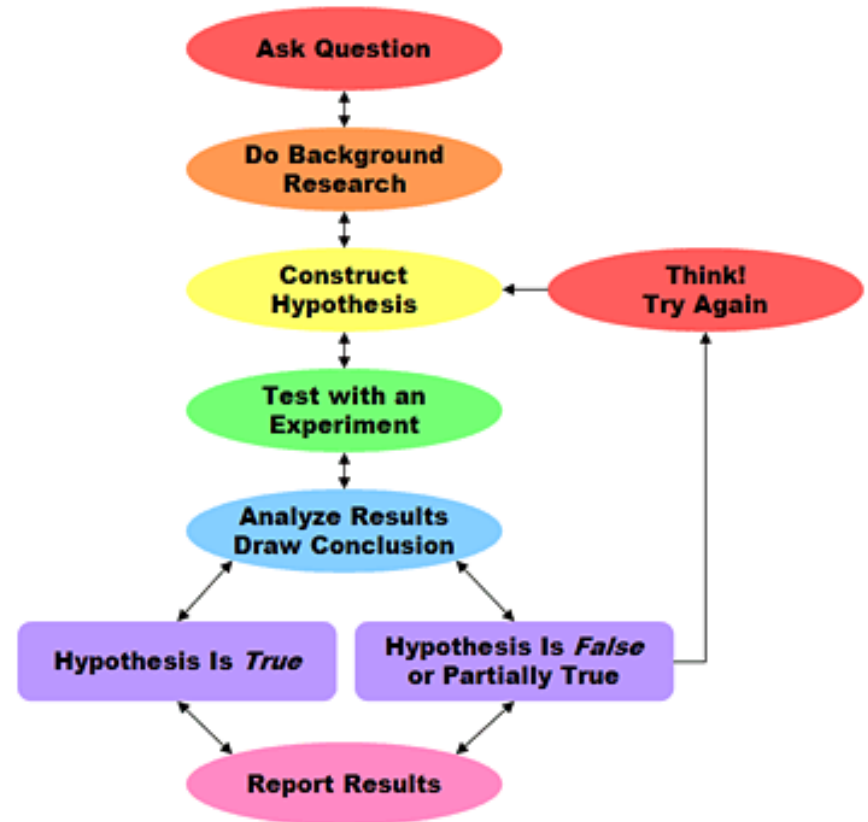
FISH 290

•The “classic” scientific method:

- Pose a null and one or more alternative hypotheses
- Conduct an experiment and record the data
- Examine the data with statistical techniques
- Reject the null hypothesis (with some risk of error),
- Or, be unable to reject the null hypothesis and so have to accept it as correct (with some risk of error).

http://www.sciencebuddies.org/science-fair-projects/project_scientific_method.shtml

Steps of the scientific method



Ask a question:

For the scientific method to answer the question it **must be about something that you can test**, preferably something that you can quantify (measure, count, etc.).

Do background research: Rather than starting from scratch in putting together a plan for answering your question, you'll want to do a library/internet/interview search to help you find the best way to do things and ensure that you aren't repeating what's already been done.



•**Construct a hypothesis:** A hypothesis is an educated guess about how things work:

•“If _____[*I do this*] _____, then _____[*this*]_____ will happen.”

•“If _[*some theory or idea*]_____ is true, then we should expect to see __[*data or action*]_____.”

•You must state your hypothesis in a way that you can **easily measure** and, of course, your hypothesis should be constructed in a way to help you **answer your original question**.



Test your hypothesis by doing an experiment

Experiments have two essential features:

1) Treatment and control

2) Replication

What about other forms of analysis and inference?

Do we always need to be *explicitly* erecting and falsifying hypotheses?

We might want to estimate the relative effect of different processes on some complex outcome, rather than narrowly determining whether each, in isolation, has an effect or not.

We might want to estimate the relative likelihood of a series of possible outcomes, rather than just stating which is the most likely to occur.

The scientific process is as diverse as the people practicing science.

Approaches that do not involve direct experimentation?

Observation (animal behavior, environmental change, etc.)

Correlation (density and growth, salinity and survival, etc.)

Modeling (using the computer to explore scenarios, options)

Morphology (paleontology, etc.)

Descriptive (discovery of a new species of organism)

Microscopical examination (new pathogens, structures, etc.)

Others

Are null hypotheses imbedded in these non-experimental studies?



Communicate your results: To complete your project you communicate your results to others.

- presenting your findings orally at a scientific meeting
 - a poster at a scientific meeting
 - publishing the work in a scientific journal
 - Publishing results in popular media
- 

Where do we find scientific information?

Peer-reviewed
journals

Single-authored
books

Edited, multi-
authored books

Master's theses
and doctoral
dissertations

Agency reports
(sometimes called
“grey literature”)

Websites and
other electronic
sources

Others?

Each form of communication has strengths and weaknesses; it is important to understand their differences.

Types of scientific literature

- **Primary:** represents the results of an original study
 - Scientific papers in IMRaD format*
 - *note that some journals like *Science* and *Nature* use a different format more like a magazine article
 - Also includes conference papers, dissertations, and technical reports
 - How do you know you're reading primary literature? Look for (mention of) Methods and Results.



Types of scientific literature

• **Primary:** represents the results of an original study

- Not all primary literature is peer-reviewed.
- **Grey literature** refers to reports that are archived but have not been peer-reviewed. Include dissertations, technical reports, and pre-prints





Types of scientific literature

- **Secondary:** sources *based* on multiple primary literature sources
 - Books
 - Popular articles
 - Review papers in scientific journals. **Review papers are peer reviewed!**
 - Great for readers to get an overview on a topic
 - Sometimes contain new insights based on surveying the literature

How is science published?

1. Choose a journal to submit your manuscript

- Often you write the draft of the manuscript with this journal in mind far in advance
- Each journal has specific formatting requirements you must follow

2. Submit the manuscript online

3. An editor sends the manuscript out to anonymous peer reviewers

- Who are peer reviewers and how are they chosen?

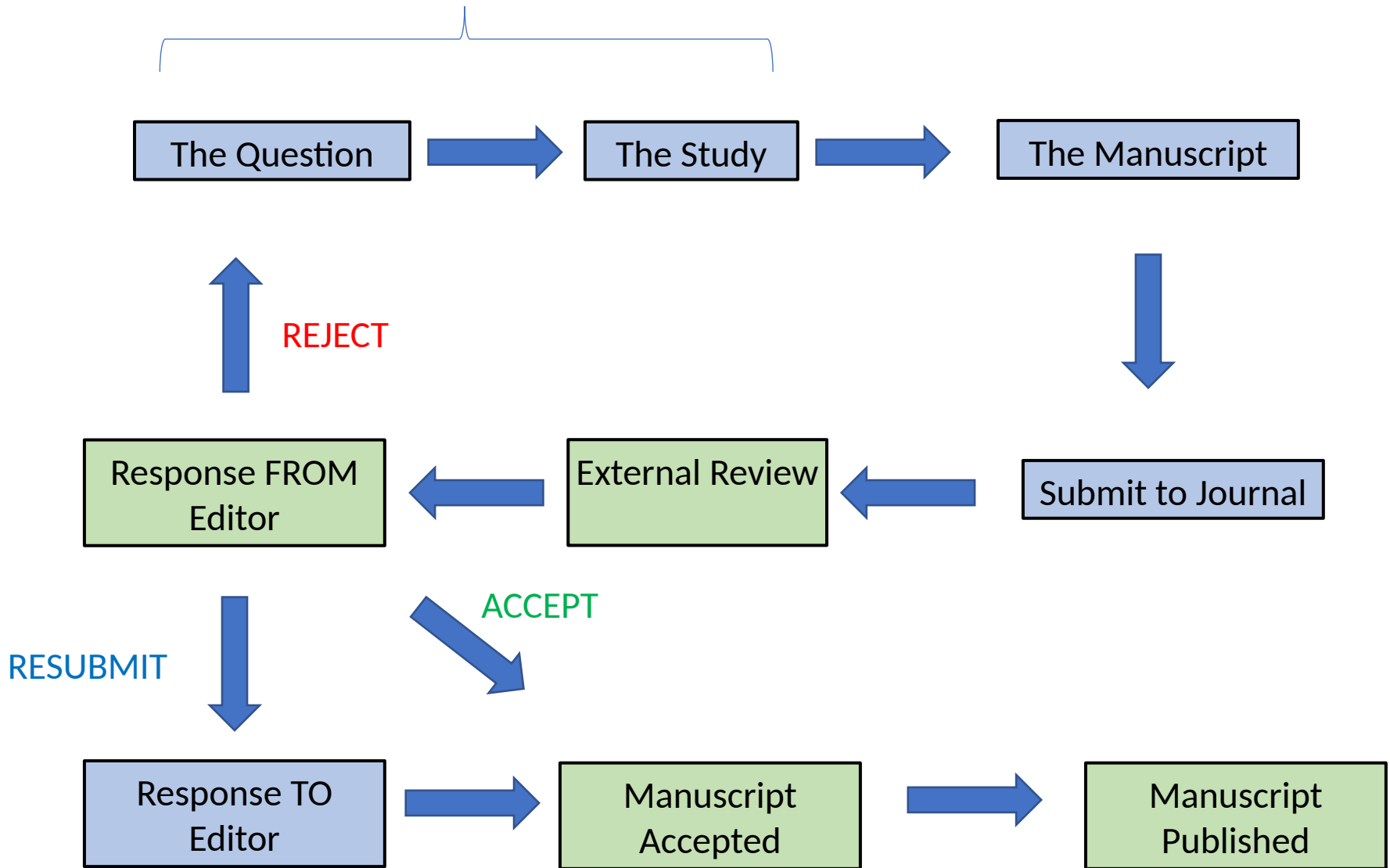
4. Reviews come back and the editor makes a decision

- Usually five options: Accept-as-is, Minor revisions, Major revisions, Reject + resubmit, or Reject without option to resubmit

5. Manuscript is accepted to the journal!

- Journal staff type-set the article and it is published (usually a month or more after acceptance)

The Scientific Method



How is science published?

An example

PROCEEDINGS B

rspb.royalsocietypublishing.org

Research



Cite this article: Miller EC, Hayashi KT, Song D, Wiens JJ. 2018 Explaining the ocean's richest biodiversity hotspot and global patterns of fish diversity. *Proc. R. Soc. B* **285**: 20181314. <http://dx.doi.org/10.1098/rspb.2018.1314>

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Explaining the ocean's richest biodiversity hotspot and global patterns of fish diversity

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For most marine organisms, species richness peaks in the Central Indo-Pacific region and declines longitudinally, a striking pattern that remains poorly understood. Here, we used phylogenetic approaches to address the causes of richness patterns among global marine regions, comparing the

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Lessons:

1. Publishing takes a long time
2. Rejection is the NORM!

(cue 5 stages of grief)

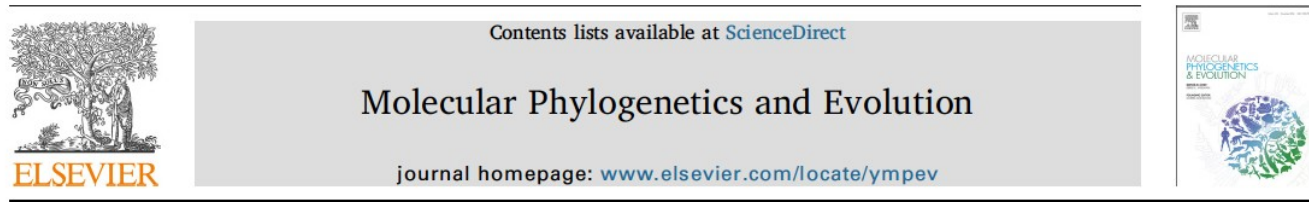
Behind the scenes, it took 1 year from final draft to published paper

- October 2017: finished the “final” draft, submit *PNAS*
- November 2017: *PNAS* **reject**, reformat + submit *PLoS Biology*.
 - 2 days later *PLoS Biol* **reject**, reformat + submit to *Proc. B*
- March 2018: Decision: **Reject** + resubmit
- June 2018: revisions done, sent back to *Proc B*
- August 2018: Decision: Minor revisions
- September 2018: revisions done, sent back to *Proc B*
- September 2018: **Accept** as-is
- October 2018: Paper comes out in journal

How is science published?

Another example:

- “Final draft” submitted to Systematic Biology – summer 2014
- 3 rejections in 3 different journals
- 2 additional rounds of revision in Molecular Phylogenetics and Evolution
- Finally published in 2018



Phylogenomic analysis on the exceptionally diverse fish clade Gobioidae (Actinopterygii: Gobiiformes) and data-filtering based on molecular clocklikeness

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Scientific journals and articles:

So much to read; so little time.

So, what are we looking for?

Class activity 1:

We will look at the names of selected journals and you will indicate whether you would regard them as

- 1) likely to read
- 2) might read if I had time
- 3) not likely to read

Remember, we are all busy and no one can read everything. There are 12 journals, so you must choose 4 to read, 4 “maybes” and 4 that you’d skip.

Would you read this journal:

<u>Journal title</u>	<u>Yes</u>	<u>Maybe</u>	<u>Not Likely</u>
Marine Biology			
Freshwater Biology			
Animal Behaviour			
Ecological Modeling			
Ecology			
Molecular Ecology			
Journal of Fish Biology			
Journal of Crustacean Biology			
Journal of Fish Diseases			
Science			
Canadian Journal of Zoology			
Canadian Journal of Fisheries and Aquatic Sciences			

Let's hear your ideas!

Let's narrow down your ideas into a question
and a set of testable hypotheses

Project titles from last quarter (and where they found their data)

- **Analysis of Invasive European Green Crab Diet and Impact on Atlantic Ocean Ecosystems** – Several unique peer reviewed studies combined
- **Niche Partitioning of Jellyfish in the Puget Sound** – dataset from Luke
- **Change in Freshwater Fish Aggression from Increased Temperature** - Several unique peer reviewed studies
- **Small body size as a contributing factor of symbiosis in Gobies and Wrasses**– literature review, Fishbase for body size
- **Fisheries Management Plan effects on Horseshoe crabs** – NOAA
- **Habitat Partitioning in Puget Sound Flatfishes** – WDFD commercial fishing data
- **Human population density impacts on freshwater macroinvertebrates** - King County Benthic Macroinvertebrate Ambient Monitoring Program
- **Comparing PCBs in Puget Sound Fishes** – WDFW/Luke
- **Top down vs Bottom up population control in three North Atlantic fishes** - NOAA

Project titles from previous quarters (and where they found their data)

- Infection Rate Factors of Stony Coral Tissue Loss Disease in the Florida Keys and Caribbean- NOAA
- Effects of Abiotic Factors on the Foraging Behavior of *Enhydra lutra*- WA state monitoring data
- Mercury Content of Puget Sound Rockfish- Puget Sound Ambient Monitoring Program
- Impact of Volcanic Activity on Phytoplankton Blooms in the Bering Sea- combo of Alaskan Volcanic Observatory and NOAA data
- Effects of physical water conditions on *Neocaridina* invasions in the rivers of Hiroshima Prefecture- data from a peer-reviewed study

Project titles from previous quarters (and where they found their data)

- Decoding the Population Decline of Dungeness Crab, *C. magister*, in South Puget Sound- WA Dept Fish and Wildlife
- Effects of Skagit River discharge on steelhead smolt outmigration- combo USGS and WDFW
- Relative Brain Size and Other Factors Contributing to Cetacean Extinction- combo IUCN and data from a peer-reviewed study
- East and West Coast Analysis of the Effectiveness of Marine Protected Areas- data from peer-reviewed studies
- Effect of *Procambarus clarkii* on Abundance of Trichopterans and Diversity of Macroinvertebrates in Pine Lake, Washington- Puget Sound Benthos Survey

For MONDAY

Continue thinking about your question and make it more focused. After thinking about sources of data and potential limitations, **write down your more focused question** and hypotheses and **be prepared to share next class.**